1. Write a program in C to create two shared libraries:

- **Library 1**: `libperfect.so`
  Contains function:
  - `int IsPerfect(int No);`
- **Library 2**: `libhelper.so`
  Contains helper function(s) used by `IsPerfect()`:
  - `int SumOfFactors(int No);` *(or)*
  - `int IsFactor(int No, int i);` *(any helper logic)*

Requirements

1. `libperfect.so` must **internally call** helper function(s) from `libhelper.so`.
2. Client application should call only `IsPerfect()` from `libperfect.so`.
3. Demonstrate linking chain:
   Client → libperfect.so → libhelper.so

Expected Output

- Input: `28` → "28 is a Perfect Number"
- Input: `12` → "12 is NOT a Perfect Number"

Deliverables

- `helper.c`, `helper.h` → `libhelper.so`
- `perfect.c`, `perfect.h` → `libperfect.so` (linked with libhelper)
- `client.c`

2. Write a program in C to create two shared libraries:

- `libaddition.so` containing:
  - `int Addition(int A, int B);`
- `libsubstraction.so` containing:
  - `int Substraction(int A, int B);`

Requirements

1. Create **separate** `.so` for each operation.
2. Client should link **both** `.so` libraries.

3. Client should accept two numbers and call both functions.

Expected Output

- Addition: `A + B`
- Substraction: `A - B`

3. Write a program in C to create a shared library `libdisplay.so` which contains:

- `void Display(char *str);`

Requirements

1. Client accepts a string from user.
2. Client calls `Display()` from shared library.
3. Library must display:
    - The string
    - Total characters (optional)

4. Write a program in C to create a shared library `libmystring.so` which implements:

- `int MyStrlen(const char *str);`

Requirements

1. Do **not** use `strlen()` inside the library.
2. Client accepts string from user and calls `MyStrlen()`.
3. Display length.

Additional Condition

- Handle `NULL` string safely.

5. Write a program in C to create two shared libraries:

1. `libmystring.so` implementing major `string.h` functions
2. `libmymath.so` implementing major `math.h` functions

Suggested Functions

`string.h` set (implement at least 8)

- `MyStrlen`

- `MyStrcpy`
- `MyStrncpy`
- `MyStrcat`
- `MyStrcmp`
- `MyStrncmp`
- `MyStrrev`
- `MyStrchr` / `MyStrstr`

`math.h` set (implement at least 8)

- `MyPow`
- `MySqrt` (approx OK)
- `MyAbs`
- `MyMax`
- `MyMin`
- `MyFactorial`
- `MyGCD`
- `MyLCM`

Requirements

1. Client should call **all functions** and display output.
2. Provide menu-driven client.
3. No usage of original library functions internally (implement your own logic).